

Take-Away and Comparing Stories

Answer Key

Kindergarten–Grade 1

Quick Answers

- | | |
|-----------------------------------|---|
| 1. 4 balloons | 6. 4 grapes |
| 2. 5 cookies | 7. 3 marbles |
| 3. 5 | 8. flowers Ava has left = 7, how many more = 2, |
| 4. 2 stickers | — |
| 5. how many extra = 2 counters, — | |
-

What is verified

6 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 5, 8. The worked solution states what a correct response must contain.

The author marked 2 blanks as working space (*not* answers — nothing checks what lands there).

Curriculum

Level: Kindergarten–Grade 1

1.OA.C.6 – *problems 4, 5, 7, 8*

K.OA.A – *problems 1, 2, 3, 6*

Difficulty 1–4 of 5 across 11 tagged checks.

- 1.** Jon has 6 balloons. 2 balloons pop. How many balloons does Jon have left?

Solution: Popping is taking away: start with 6, take away 2. Count back from 6: 5, 4. So $6 - 2 = 4$. 4 balloons

- 2.** There are 8 cookies on a plate. Max eats 3 cookies. Draw the cookies and cross out the eaten ones if it helps. How many cookies are left?

Solution: Draw 8 cookies and cross out 3 for the ones Max ate. Counting the plain cookies: 5. So $8 - 3 = 5$. 5 cookies

- 3.** 9 kites are flying. 4 kites come down to land. Write the story numbers in the equation, then write how many kites are still flying.

Solution: Landing takes kites out of the sky: the start number is 9 and the taken-away number is 4, so the equation is $9 - 4 = ?$ Count back from 9: 8, 7, 6, 5. So $9 - 4 = 5$. 5

- 4.** Lena has 7 stickers. Ben has 5 stickers. How many more stickers does Lena have than Ben?

Solution: Compare by matching: pair each of Ben's 5 stickers with one of Lena's. Lena's extras are $7 - 5 = 2$. 2 stickers

5. Row A shows 6 counters. Row B shows 4 counters. Draw a line from each counter in Row A to a partner in Row B. How many counters in Row A have no partner?

Solution: Each of Row B's 4 counters pairs with one counter in Row A, so 4 counters get partners and the rest do not: $6 - 4 = 2$. 2 counters

Matching lines (instructor judges): 4 pairing lines drawn, leaving exactly 2 counters in Row A unmatched.

6. Zoe has 10 grapes. She eats 4 grapes, and then she eats 2 more. How many grapes does Zoe have left?

Solution: Take away twice: $10 - 4 = 6$ after the first snack, then $6 - 2 = 4$ after the second. So $10 - 4 - 2 = 4$. 4 grapes

7. Sam has 9 marbles. Tia has 6 marbles. How many fewer marbles does Tia have than Sam?

Solution: Match Tia's 6 marbles against Sam's 9. Sam has $9 - 6 = 3$ with no partner, so Tia has 3 fewer. 3 marbles

8. Challenge! Ava picks 10 flowers. She gives 3 flowers to her mom. Kim has 5 flowers. How many flowers does Ava have left? Who has more flowers now? How many more?

Solution: First the take-away step: Ava keeps $10 - 3 = 7$ flowers. 7

Now compare: Ava's 7 against Kim's 5. Matching them up leaves $7 - 5 = 2$ extras for Ava, so Ava has more — by 2

Who has more (instructor judges): the student writes "Ava", matching the counts 7 and 5.